WoSM Academy

M201-F27 · Calculus II

Infinite series

Geometric and telescoping series

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WoSM Academy

Lecture 01 · Unit 1 · Fall 2027

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Calculus II at a glance

25 units in four parts

- Units 1–8: series, power series, Taylor series
- Units 9–17: differential calculus in $\mathbb{R}^n$ and optimisation
- Units 18–22: multiple integrals
- Units 23–25: ordinary differential equations

Assessment

- Unit quizzes, best 10 of 12: 20%
- Project 1, series and generating functions: 15%
- Project 2, optimisation in several variables: 20%
- Final exam: 45%

Prerequisites: Calculus I (M102-S27) and Linear Algebra (M103-S27). We use limits of sequences, improper integrals and the bounds $\int f \le \sum f(k)$ from Calculus I throughout this part.

This lecture

01 Adding infinitely many numbers

02 Series and partial sums

03 Geometric series

04 First properties

05 Telescoping series

06 The harmonic series

07 Summary and practice

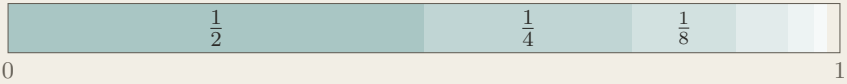
Reading: G. Strang, E. Herman, *Calculus, Volume 2* (OpenStax), Section 5.2.

01

Adding infinitely many numbers

Can we add infinitely many numbers?

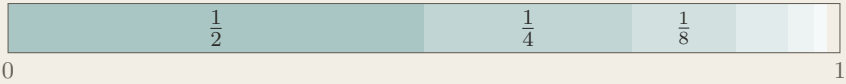
Walk halfway to a wall, then half of the remaining distance, then half again, …



After n steps you have covered $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$, and $\frac{1}{2^n}$ is left.

Can we add infinitely many numbers?

Walk halfway to a wall, then half of the remaining distance, then half again, …



After n steps you have covered $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$, and $\frac{1}{2^n}$ is left.

It seems right to write

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1.$$

But the additions never finish. What does “=” mean here?

Three answers to one question

Let $S = 1 - 1 + 1 - 1 + 1 - 1 + \cdots$. Three "proofs":

$$S = (1 - 1) + (1 - 1) + (1 - 1) + \cdots = 0 + 0 + 0 + \cdots = 0$$

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$$S = 1 - (1 - 1 + 1 - 1 + \cdots) = 1 - S \quad \Longrightarrow \quad S = \tfrac{1}{2}$$

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The lesson

The rules for finite sums — regrouping, reordering, solving for the sum — do not automatically hold for infinite sums. We need a **definition** of an infinite sum, and then **theorems** that say which manipulations are allowed.

Where series appear in this module and later

- **Functions as series** (Units 5–6): $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, $\quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$
- **Counting and recurrences** (Unit 7, Project 1): generating functions
- **Approximation** (Unit 8): Stirling's formula for $n!$
- **Algorithms**: the cost of a growing array, recursion trees, the $\ln n$ in the analysis of Quicksort
- **Probability**: the probabilities of infinitely many outcomes add up to 1; expected values are series

Today: what a convergent series is, the two families of series we can sum exactly (geometric and telescoping), and the first warning sign of divergence.

02

Series and partial sums

Recall: sequences (Calculus I)

- (s_n) **converges** to s if for every $\varepsilon > 0$ there is an N such that $|s_n - s| < \varepsilon$ for all $n \geq N$.
- **Limit laws**: if $s_n \to s$ and $t_n \to t$, then $s_n + t_n \to s + t$, $\; c\,s_n \to c\,s$, $\; s_n t_n \to st$.
- **Monotone convergence**: an increasing sequence that is bounded above converges (to its supremum). An increasing sequence that is not bounded above tends to ∞.
- **Cauchy criterion**: (s_n) converges if and only if for every $\varepsilon > 0$ there is an N with $|s_n - s_m| < \varepsilon$ for all $n, m \geq N$.
- If $|r| < 1$, then $r^n \to 0$.

Series

Definition (Series, partial sums, sum)

Let $(a_k)_{k\ge 1}$ be a sequence of real numbers. Its n-th **partial sum** is

$$S_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k.$$

The **series** $\sum_{k=1}^{\infty} a_k$ **converges** if the sequence (S_n) converges, and then its **sum** is $\displaystyle\sum_{k=1}^{\infty} a_k = \lim_{n\to\infty} S_n$. Otherwise the series **diverges**.

If $S_n \to \infty$ we write $\sum_{k=1}^{\infty} a_k = \infty$ and say the series **diverges to** ∞. A series *is* its sequence of partial sums, so everything we know about sequences applies.

First examples

Example (Halves)

$a_k = \frac{1}{2^k}$: $S_1 = \frac{1}{2},\ S_2 = \frac{3}{4},\ S_3 = \frac{7}{8},\ \ldots,\ S_n = 1 - \frac{1}{2^n} \to 1,$ so $\sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$

First examples

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Example (Grandi's series)

$a_k = (-1)^{k+1}$: $\quad S_1 = 1,\ S_2 = 0,\ S_3 = 1,\ S_4 = 0,\ \ldots$ has no limit, so the series diverges.

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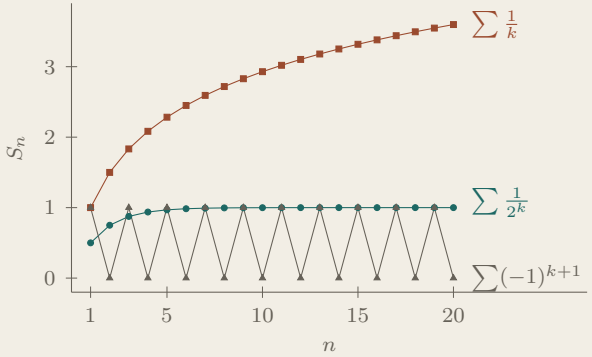
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Example (Natural numbers)

$a_k = k$: $S_n = \frac{n(n+1)}{2} \to \infty,$ so $\sum_{k=1}^{\infty} k = \infty.$

Partial sums, plotted



- $\sum 1/2^k$: the partial sums settle at 1.
- $\sum(-1)^{k+1}$: they jump between 1 and 0 forever.
- $\sum 1/k$: they keep growing, ever more slowly. Do they settle? (Section 6.)

Notation and the starting index

- A series may start at any index: $\displaystyle\sum_{k=0}^{\infty} r^k$, $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k^2-1}$. Partial sums then start there too.
- Shifting the index changes nothing: $\displaystyle\sum_{k=1}^{\infty} a_k = \sum_{j=0}^{\infty} a_{j+1}$ (put $j = k-1$).
- The name of the index does not matter: $\sum_k a_k = \sum_j a_j$.
- When the range is clear we write $\sum a_k$.

Warning

The symbol $\sum_{k=1}^{\infty} a_k$ stands for two things: the series (the sequence of partial sums) and, if it converges, its sum (a number). Writing $\sum_{k=1}^{\infty} a_k = S$ *claims* that the series converges.

03

Geometric series

Finite geometric sums

Lemma

For $r \neq 1$ and $n \geq 0$:

$$\sum_{k=0}^{n} r^k = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Finite geometric sums

Lemma

For $r \neq 1$ and $n \geq 0$:

$$\sum_{k=0}^{n} r^k = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Proof.

Let $G_n = 1 + r + \cdots + r^n$. Then $rG_n = r + r^2 + \cdots + r^{n+1}$, so in $G_n - rG_n = 1 - r^{n+1}$ every other term cancels — our first *telescoping* sum (Section 5). Divide by $1 - r \neq 0$. □

The geometric series

Theorem (Geometric series)

$\sum_{k=0}^{\infty} r^k$ converges if and only if $|r| < 1$, and then $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$.

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Proof.

Use $G_n = \frac{1-r^{n+1}}{1-r}$ for $r \neq 1$.

- $|r| < 1$: $r^{n+1} \to 0$, so $G_n \to \frac{1}{1-r}$.
- $r = 1$: $G_n = n + 1 \to \infty$. $r = -1$: $G_n = 1, 0, 1, 0, \ldots$ has no limit.
- $|r| > 1$: $|G_n| \geq \frac{|r|^{n+1}-1}{|1-r|} \to \infty$: unbounded, so divergent. □

First term over one minus the ratio

Corollary

If $|r| < 1$, then for any number a and any integer m

$$\sum_{k=m}^{\infty} ar^k = \frac{ar^m}{1-r} = \frac{\text{first term}}{1-\text{ratio}}.$$

Proof: the partial sums are $ar^m G_n$ (put $j = k - m$); use the limit laws.

To use it: identify the first term and the ratio r, and check that $|r| < 1$.

Examples

$$\sum_{k=1}^{\infty} \frac{3}{4^k}$$

$$\sum_{k=0}^{\infty} \left(-\frac{2}{3}\right)^k$$

$$\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k}$$

$$\sum_{k=0}^{\infty} \left(\frac{3}{2}\right)^k$$

Examples

$$\sum_{k=1}^{\infty} \frac{3}{4^k} = \frac{3/4}{1 - 1/4} = 1$$

$$\sum_{k=0}^{\infty} \left(-\frac{2}{3}\right)^k = \frac{1}{1 + 2/3} = \frac{3}{5}$$

$$\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k} = \frac{8/25}{1 - 2/5} = \frac{8}{15}$$

$$\sum_{k=0}^{\infty} \left(\frac{3}{2}\right)^k \text{ diverges: } |r| = \tfrac{3}{2} \geq 1$$

Examples

$$\sum_{k=1}^{\infty} \frac{3}{4^k} = \frac{3/4}{1 - 1/4} = 1 \qquad \sum_{k=0}^{\infty} \left(-\frac{2}{3}\right)^k = \frac{1}{1 + 2/3} = \frac{3}{5}$$

$$\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k} = \frac{8/25}{1 - 2/5} = \frac{8}{15} \qquad \sum_{k=0}^{\infty} \left(\frac{3}{2}\right)^k \text{ diverges: } |r| = \tfrac{3}{2} \geq 1$$

Common mistakes

The first term is not always the coefficient: in $\sum_{k=2}^{\infty} \frac{2^{k+1}}{5^k}$ it is $\frac{2^3}{5^2} = \frac{8}{25}$. And $\frac{1}{1-r}$ is the sum only when the series starts at $r^0 = 1$.

Repeating decimals

Example

$$0.999\ldots = \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \cdots = \frac{9/10}{1 - 1/10} = 1.$$

- The same argument shows that every repeating decimal is a rational number.
- A decimal expansion *is* a series: $0.d_1d_2d_3\ldots = \sum_{k=1}^{\infty} d_k\, 10^{-k}$.
- In binary, $0.0111\ldots_2 = \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1}{2} = 0.1_2$: some numbers have two expansions.

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$$0.363636\ldots = \frac{36}{100} + \frac{36}{100^2} + \frac{36}{100^3} + \cdots = \frac{36/100}{1 - 1/100} = \frac{36}{99} = \frac{4}{11}.$$

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Application: an array that doubles

When a growable array is full, a common implementation allocates one of twice the capacity and copies every element across: capacities $1, 2, 4, 8, \ldots$

- Inserting n elements copies $1, 2, 4, \ldots, 2^j$ elements, where 2^j is the largest power of 2 below n. In total, reading from the largest term:

$$2^j\Big(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^j}\Big) < 2^j \sum_{i=0}^{\infty} \frac{1}{2^i} = 2 \cdot 2^j < 2n.$$

  So n insertions cost $O(n)$ in total: $O(1)$ per insertion, amortised.

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  So n insertions cost $O(n)$ in total: $O(1)$ per insertion, amortised.

> **Remark**
>
> Terms that shrink geometrically sum to at most a constant times the largest one. So $T(n) = T(n/2) + n$ gives $T(n) \le n\,(1 + \frac{1}{2} + \frac{1}{4} + \cdots) = 2n$.

Application: waiting for a success

Repeat independent trials, each succeeding with probability p, $0 < p < 1$.

$$P(\text{first success at trial } k) = (1-p)^{k-1}\,p, \qquad k = 1, 2, 3, \ldots$$

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These probabilities form a geometric series with ratio $1 - p$, and $|1 - p| < 1$:

$$\sum_{k=1}^{\infty} (1-p)^{k-1} p = \frac{p}{1-(1-p)} = 1.$$

So, with probability 1, a success eventually happens.

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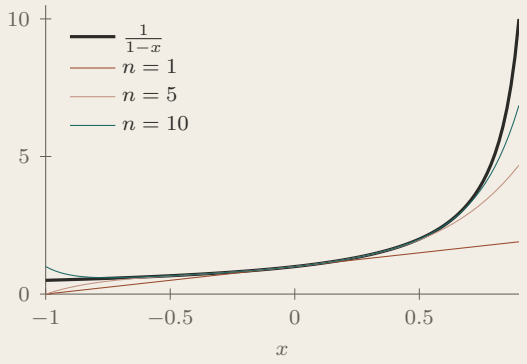
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> **Later in the module**
>
> The expected number of trials is $\sum_{k=1}^{\infty} k\,(1-p)^{k-1}p = \frac{1}{p}$. Summing it needs $\sum_{k\geq 1} kx^{k-1} = \frac{1}{(1-x)^2}$, which we prove in Unit 5 by differentiating a power series.

A first power series



For $-1 < x < 1$:

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$$

and the series diverges for $|x| \geq 1$.

The partial sums $1 + x + \cdots + x^n$ approach $\frac{1}{1-x}$ for every x in $(-1, 1)$ — more slowly the closer x is to ± 1. A function written as a series in powers of x is a *power series*: Unit 5.

04

First properties

Linearity

Theorem

If $\sum a_k = A$ and $\sum b_k = B$ both converge and $c \in \mathbb{R}$, then

$$\sum (a_k + b_k) = A + B \qquad \text{and} \qquad \sum c\, a_k = cA.$$

Proof: the partial sums are $A_n + B_n$ and cA_n; use the limit laws.

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Proof: the partial sums are $A_n + B_n$ and cA_n; use the limit laws.

Example

$$\sum_{k=0}^{\infty} \frac{2^k + 3^k}{6^k} = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^k + \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{3}{2} + 2 = \frac{7}{2}.$$

Splitting a series

> **Warning**
>
> Linearity works one way only: $\sum(1 + (-1)) = 0$ converges, but $\sum 1$ and $\sum(-1)$ diverge. Split a series only when you know both parts converge.

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Corollary

If $\sum a_k$ converges and $\sum b_k$ diverges, then $\sum(a_k+b_k)$ diverges.

Proof.

If $\sum(a_k+b_k)$ converged, then by linearity so would $\sum((a_k+b_k)-a_k)=\sum b_k$. □

For example, $\sum\left(\frac{1}{2^k}+1\right)$ diverges.

Only the tail matters

Theorem

Let $m \geq 1$. Then $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=m+1}^{\infty} a_k$ converges, and then

$$\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{m} a_k + \sum_{k=m+1}^{\infty} a_k.$$

Proof: for $n > m$, $S_n = S_m + \sum_{k=m+1}^{n} a_k$, and S_m is a fixed number.

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Proof: for $n > m$, $S_n = S_m + \sum_{k=m+1}^{n} a_k$, and S_m is a fixed number.

- Changing, adding or removing *finitely many* terms does not change whether a series converges. It may change the sum.
- If $\sum a_k = S$, the **remainder** $R_n = \sum_{k=n+1}^{\infty} a_k = S - S_n$ tends to 0. The error of stopping after n terms is R_n.

The n-th term test

Theorem

If $\sum a_k$ converges, then $a_k \to 0$.

Proof.

$a_n = S_n - S_{n-1} \to S - S = 0.$ □

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Corollary (Divergence test)

If (a_k) does not converge to 0, then $\sum a_k$ diverges.

Using the n-th term test

Example

These all diverge, because their terms do not tend to 0:

- $\sum \frac{k}{k+1}$ (terms $\to 1$) $\qquad$ $\sum k \sin \frac{1}{k}$ (terms $\to 1$)
- $\sum (-1)^k$ (terms have no limit) $\qquad$ $\sum r^k$ with $|r| \geq 1$ ($|r^k| \geq 1$)

Using the n-th term test

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Warning

The converse is false: $a_k \to 0$ does *not* imply that $\sum a_k$ converges (Sections 5 and 6). The test can only prove divergence.

Grouping terms

Theorem

If $\sum a_k$ converges to S, then inserting parentheses — grouping consecutive terms into blocks — gives a series that also converges to S.

Proof: the partial sums of the grouped series are $S_{n_1}, S_{n_2}, S_{n_3}, \dots$ for some $n_1 < n_2 < \cdots$: a subsequence of (S_n), which has the same limit.

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Removing parentheses is not allowed: $(1-1)+(1-1)+\cdots = 0$ converges, but $1-1+1-1+\cdots$ diverges.

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So for $1-1+1-\cdots$: the first two "proofs" grouped a *divergent* series, and the third assumed that the sum exists. None of 0, 1, $\frac{1}{2}$ is its sum.

The Cauchy criterion

Theorem (Cauchy criterion for series)

$\sum a_k$ converges if and only if for every $\varepsilon > 0$ there is an N such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon \qquad \text{for all } n > m \geq N.$$

Proof.

$a_{m+1} + \cdots + a_n = S_n - S_m$, and (S_n) converges if and only if it is a Cauchy sequence (Calculus I, Unit 4). □

- It proves convergence *without knowing the sum.* Lecture 3 uses it for absolutely convergent series.
- With $n = m + 1$ it gives the n-th term test again.

05

Telescoping series

Telescoping sums

If each term is a difference of consecutive terms of another sequence, $a_k = b_k - b_{k+1}$, the partial sums collapse:

$$S_n = (b_1 - b_2) + (b_2 - b_3) + \cdots + (b_n - b_{n+1}) = b_1 - b_{n+1}.$$

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Theorem (Telescoping series)

$\sum_{k=1}^{\infty}(b_k - b_{k+1})$ converges if and only if (b_n) converges, and then

$$\sum_{k=1}^{\infty}(b_k - b_{k+1}) = b_1 - \lim_{n\to\infty} b_n.$$

Proof: $S_n = b_1 - b_{n+1}$. The finite geometric sum used the same cancellation.

Example: $\sum \frac{1}{k(k+1)}$

Partial fractions (Calculus I): $\displaystyle \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, so $b_k = \frac{1}{k}$.

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and in general $S_n = 1 - \frac{1}{n+1} \to 1$.

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$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots = 1.$$

We found the sum because S_n has a closed form. Usually it does not, and we can only decide *whether* a series converges (Lecture 2).

Example: a gap of two

$$\sum_{k=2}^{\infty} \frac{1}{k^2-1}, \qquad \frac{1}{k^2-1} = \frac{1}{(k-1)(k+1)} = \frac{1}{2}\Big(\frac{1}{k-1} - \frac{1}{k+1}\Big).$$

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In $\sum_{k=2}^{n}\big(\frac{1}{k-1} - \frac{1}{k+1}\big)$ the positive terms are $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-1}$ and the negative terms are $\frac{1}{3}, \dots, \frac{1}{n-1}, \frac{1}{n}, \frac{1}{n+1}$. Two survive at each end:

$$S_n = \frac{1}{2}\Big(1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}\Big) \longrightarrow \frac{1}{2}\cdot\frac{3}{2} = \frac{3}{4}.$$

Example: three factors

$$\frac{1}{k(k+1)(k+2)} = \frac{1}{2}\Big(\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)}\Big)$$

(check: the bracket is $\frac{(k+2)-k}{k(k+1)(k+2)}$).

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(check: the bracket is $\frac{(k+2)-k}{k(k+1)(k+2)}$). So the term is $b_k - b_{k+1}$ with $b_k = \frac{1}{2k(k+1)}$, and $b_n \to 0$:

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)(k+2)} = b_1 - \lim_{n\to\infty} b_n = \frac{1}{4}.$$

Example: telescoping series that diverge

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$$\sum_{k=1}^{\infty} \ln\Big(1+\frac{1}{k}\Big) = \sum_{k=1}^{\infty} \big(\ln(k+1) - \ln k\big): \qquad S_n = \ln(n+1) - \ln 1 = \ln(n+1) \to \infty.$$

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$$\sum_{k=1}^{\infty} (\sqrt{k+1} - \sqrt{k}): \qquad S_n = \sqrt{n+1} - 1 \to \infty.$$

In both the terms tend to 0: $\ln(1+\frac{1}{k}) \to 0$ and $\sqrt{k+1} - \sqrt{k} = \frac{1}{\sqrt{k+1}+\sqrt{k}} \to 0$.
They are our first divergent series with $a_k \to 0$.

Always work with partial sums

Two wrong arguments

- $\sum\left(\frac{1}{k}-\frac{1}{k+1}\right)=\sum\frac{1}{k}-\sum\frac{1}{k+1}$: both series on the right diverge (Section 6), and $\infty-\infty$ means nothing. Linearity needs both parts to converge.
- "In $\sum\big(\ln(k+1)-\ln k\big)$ every term cancels, so the sum is $-\ln 1=0$." In fact $S_n=\ln(n+1)\to\infty$.

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- "In $\sum\big(\ln(k+1) - \ln k\big)$ every term cancels, so the sum is $-\ln 1 = 0$." In fact $S_n = \ln(n+1) \to \infty$.

Recipe

1. Write $a_k = b_k - b_{k+1}$ (partial fractions, rules for logarithms, rationalising, …).
2. Compute S_n exactly: which terms survive?
3. Take the limit of S_n.

06

The harmonic series

The harmonic series diverges

The n-th **harmonic number** is $H_n = \sum_{k=1}^{n} \frac{1}{k}$.

> **Theorem**
>
> $$\sum_{k=1}^{\infty} \frac{1}{k} = \infty.$$

The terms tend to 0, so the n-th term test says nothing. The partial sums grow (Section 2's plot), but do they stay below some bound? Two proofs that they do not.

Oresme's proof (14th century)

Proof.

Group the terms in blocks that end at powers of 2:

$$1 + \tfrac{1}{2} + \underbrace{\left(\tfrac{1}{3} + \tfrac{1}{4}\right)}_{\geq 2\cdot\frac{1}{4}} + \underbrace{\left(\tfrac{1}{5} + \cdots + \tfrac{1}{8}\right)}_{\geq 4\cdot\frac{1}{8}} + \underbrace{\left(\tfrac{1}{9} + \cdots + \tfrac{1}{16}\right)}_{\geq 8\cdot\frac{1}{16}} + \cdots$$

The block that ends at $\frac{1}{2^j}$ has 2^{j-1} terms, each at least $\frac{1}{2^j}$, so it adds at least $\frac{1}{2}$. Hence $H_{2^m} \geq 1 + \frac{m}{2} \to \infty$. As (H_n) is increasing, $H_n \to \infty$. □

A second proof

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For $x > -1$, $\ln(1+x) \le x$ (the graph of $\ln(1+x)$ lies below its tangent at 0). With $x = \frac{1}{k}$:

$$H_n = \sum_{k=1}^{n} \frac{1}{k} \;\ge\; \sum_{k=1}^{n} \ln\!\Big(1 + \frac{1}{k}\Big) = \ln(n+1) \;\to\; \infty,$$

using the telescoping sum from Section 5.

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using the telescoping sum from Section 5.

> **Remark**
>
> We compared $\sum \frac{1}{k}$ with a series that we already knew diverges, term by term. Lecture 2 turns this idea into a test: the *comparison test*.

How fast does H_n grow?

From Calculus I (Unit 26):

$$\ln(n+1) \le H_n \le 1 + \ln n.$$

n	$\ln(n+1)$	H_n	$1+\ln n$
10	2.398	2.929	3.303
100	4.615	5.187	5.605
1000	6.909	7.485	7.908
10^6	13.816	14.393	14.816

- H_n grows like $\ln n$: without bound, but very slowly.
- The first n with $H_n > 10$ is $n = 12\,367$.
- In fact $H_n - \ln n \to \gamma \approx 0.5772$, the Euler–Mascheroni constant.

Harmonic numbers in algorithms

- **Quicksort** with a uniformly random pivot, on n distinct keys, makes on average

$$2(n+1)H_n - 4n \;\approx\; 2n\ln n \quad \text{comparisons.}$$

- **Coupon collector**: draw uniformly at random, with replacement, from n coupons. The expected number of draws until every coupon has been seen is

$$nH_n \;\approx\; n\ln n.$$

In both, the logarithm comes from a harmonic sum $\sum_{k=1}^{n} \frac{1}{k}$.

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$$S_n = 1 + \sum_{k=2}^{n} \frac{1}{k^2} < 1 + \sum_{k=2}^{n} \Big(\frac{1}{k-1} - \frac{1}{k} \Big) = 2 - \frac{1}{n} < 2.$$

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An increasing sequence that is bounded above converges, so $\sum \frac{1}{k^2}$ converges, and its sum is at most 2.

> **Remark**
>
> Euler found the sum: $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \approx 1.6449$ (the Basel problem). Our argument proves convergence without finding the sum.

Looking ahead

In both $\sum \frac{1}{k}$ and $\sum \frac{1}{k^2}$ the terms tend to 0, yet one diverges and the other converges. Where is the boundary?

$$\sum \frac{1}{k^{1.01}}\,? \qquad \sum \frac{1}{\sqrt{k}}\,? \qquad \sum \frac{1}{k \ln k}\,? \qquad \sum \frac{k^{10}}{2^k}\,? \qquad \sum \frac{k!}{k^k}\,?$$

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The key fact for Lecture 2

If $a_k \geq 0$ for all k, the partial sums increase, so

$$\sum a_k \text{ converges} \iff (S_n) \text{ is bounded above.}$$

Lecture 2 turns this into practical tests: comparison, integral, ratio and root.

07

Summary and practice

Summary

- $\sum a_k$ **converges** if its partial sums S_n converge; the sum is $\lim S_n$.
- **Geometric**: $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ if $|r| < 1$, and it diverges if $|r| \geq 1$. In general: first term $/(1 - \text{ratio})$.
- **Linearity** holds for convergent series. Finitely many terms never decide convergence.
- n**-th term test**: if $\sum a_k$ converges, then $a_k \to 0$. It can only prove divergence.
- **Grouping** a convergent series is allowed; removing parentheses is not.
- **Telescoping**: $\sum(b_k - b_{k+1}) = b_1 - \lim b_n$, if the limit exists.
- **Harmonic series**: $\sum \frac{1}{k} = \infty$, with $H_n \approx \ln n$; but $\sum \frac{1}{k^2}$ converges (bounded partial sums).

Practice

1. Find $\displaystyle\sum_{k=1}^{\infty}(-1)^{k+1}\frac{5}{3^k}$.
2. Write $0.142857142857\ldots$ as a fraction.
3. Find $\displaystyle\sum_{k=1}^{\infty}\frac{1}{4k^2-1}$.
4. Find $\displaystyle\sum_{k=1}^{\infty}\ln\frac{k(k+2)}{(k+1)^2}$.

Practice, continued

5. Do these converge? (a) $\displaystyle\sum \frac{k^2}{2k^2+1}$ (b) $\displaystyle\sum \Big(1+\frac{1}{k}\Big)^{k}$
6. For which x does $\sum_{k=0}^{\infty}(x-1)^k$ converge? Find its sum.
7. True or false: if $\sum a_k$ and $\sum b_k$ both diverge, then $\sum (a_k + b_k)$ diverges.
8. Show that if $\sum a_k$ converges, then $\sum_{k=n+1}^{\infty} a_k \to 0$ as $n \to \infty$.
9. A ball is dropped from a height of $2\,\text{m}$. After each bounce it rises to $\frac{3}{5}$ of its previous height. How far does it travel in total?

Answers

1. $\frac{5}{4}$ (first term $\frac{5}{3}$, ratio $-\frac{1}{3}$).
2. $\frac{142857}{999999} = \frac{1}{7}$.
3. $\frac{1}{2}$, since $\frac{1}{4k^2-1} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$.
4. $-\ln 2$, with $b_k = \ln \frac{k}{k+1}$: the term is $b_k - b_{k+1}$, $b_1 = -\ln 2$, $b_n \to 0$.
5. Both diverge: the terms tend to $\frac{1}{2}$ and to e.
6. For $0 < x < 2$ (ratio $x - 1$); the sum is $\frac{1}{2-x}$.
7. False: $a_k = 1$, $b_k = -1$.
8. The tail is $S - S_n$, and $S_n \to S$.
9. $$2 + 2 \cdot \frac{2 \cdot \frac{3}{5}}{1 - \frac{3}{5}} = 2 + 6 = 8\,\text{m}.$$

Next time

Lecture 02 — Convergence tests (Unit 2)

- Series with nonnegative terms
- The integral test and p-series
- Comparison and limit comparison
- The ratio and root tests

Reading: Strang & Herman, *Calculus, Volume 2*, Sections 5.3, 5.4 and 5.6.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.